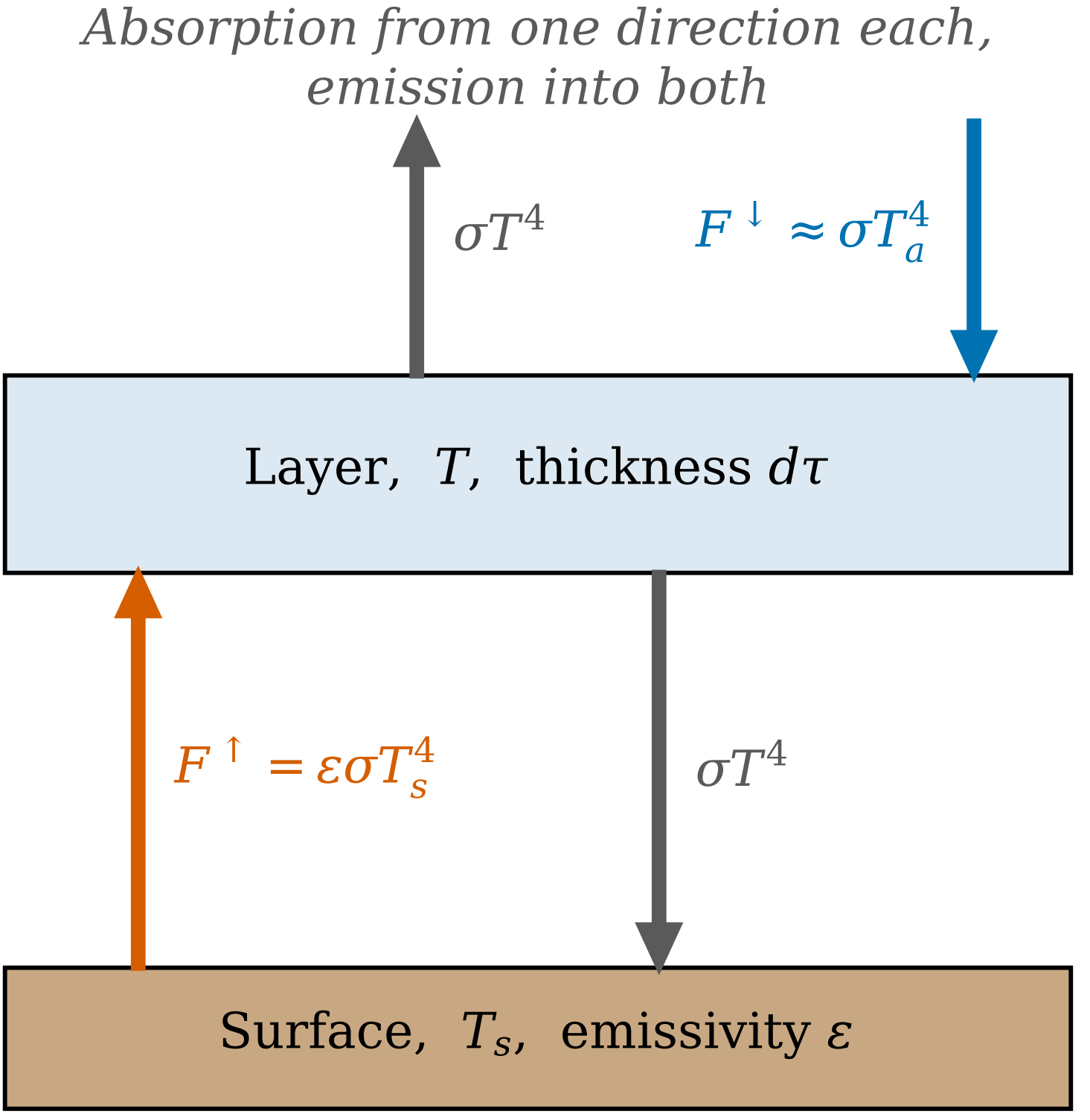

$$\frac{dE}{dt} = d\tau(F^{\uparrow} + F^{\downarrow} - 2\sigma T^4)$$